

BLOCK 1

THEORY OF INTERNATIONAL TRADE

Unit 1	Classical and Neo-Classical Theories of International Trade
Unit 2	Gains from Trade

BLOCK INTRODUCTION

This block reveals the rationale behind International trade and why countries engage in trade. It presents a comparative picture of various theories of International trade. It highlights the various gains associated with international trade and how they can be measured. A comparative picture of various types of gains has also been presented in this block.

This block consists of 2 units.

Unit 1: Classical and Neo-Classical Theories of International Trade discusses the major theories of international trade. It presents the rationale behind why countries trade with each other. The unit covers discussion on major theories, like absolute advantage theory, comparative theory, and H-O theory. Further, the theories like Stolper – Samuelson theorem, Factor Price Equalisation theory, and Rybczynski theorem developed by extending the H-O model have also been discussed.

Unit 2: Gains from International Trade throws light on sources of gains, factors determining gains from trade and how these gains can be measured. It also discusses the concept of the Production Possibilities Curve in International trade. The unit also elucidates the situation of the potential and actual gain, Free Trade versus no trade and static and dynamic gains.

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UNIT 1 CLASSICAL AND NEO-CLASSICAL THEORIES OF INTERNATIONAL TRADE*

Structure

- 1.0 Objectives
- 1.1 Introduction
- 1.2 Theory of Mercantilism
- 1.3 Absolute Advantage Theory
- 1.4 Comparative Advantage Theory
- 1.5 Heckscher–Ohlin Theory
- 1.6 Stolper – Samuelson Theorem
- 1.7 Factor-Price Equalization Theorem
- 1.8 Rybczynski Theorem
- 1.9 Lets Sum Up
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- 1.11 Some useful References
- 1.12 Answers/Hints to Check Your Progress Exercise

1.0 OBJECTIVES

After studying this Unit, you should be able to:

- Explain the difference between Domestic and International Trade;
- describe why two nations trade with each other;
- explain Absolute Advantage Theory;
- explain Comparative Advantage Theory;
- differentiate between Absolute Advantage Theory and Comparative Advantage Theory;
- describe Heckscher–Ohlin Theory;
- explain the Stolper – Samuelson theorem;
- explain Factor Price Equalisation theory; and
- describe Rybczynski Theorem.

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1.1 INTRODUCTION

In this unit, we will discuss the major theories of international trade. At the beginning of this unit, we will try to understand what international trade is and how it differs from domestic trade. The reasons why countries trade with each other will also be discussed. Having understood these, the major theories of international trade that have evolved, like absolute advantage theory, comparative theory, and H-O theory, will be explained. Further, the theories like Stolper – Samuelson theorem, Factor Price Equalisation theory, and Rybczynski theorem developed by extending the H-O model will also be discussed.

Before reading the theories of international trade you should understand what is, international trade, and why does trade take place?

In simple words, trade refers to buying and selling goods and services in exchange for money. The sellers sell the goods and services while the buyers buy the same. When these activities occur domestically in the country, it is referred to as domestic trade or only trade. But, when the exchange of goods and services takes place between buyer and seller of two different nations it is called international trade.

Now, the question is, why do we trade at all?

The simple answer to this question is that any nation cannot produce everything it needs. One nation may have plenty of natural resources but a scarcity of meat, fish, etc. One country may be an efficient milk producer but have a scarcity of wheat and rice. In this situation, nations come together and exchange goods among them. We will discuss more about gains from trade in Unit 2.

1.2 THEORY OF MERCANTILISM

Mercantilism is an exercise that is more than 500 years old. The base of this theory was the "commercial revolution", the transition from local economies to national economies, from feudalism to capitalism, and from a rudimentary trade to a larger international trade.

The theory of mercantilism postulates that countries should encourage exports and discourage imports. The tendency to export more and import less and receive gold (as gold was the medium of exchange) in exchange is called **Mercantilism**. Mercantilism was the economic system of the major trading nations during the 16th to 18th centuries. The theory assumed that national wealth and power were best served by increasing exports and collecting precious metals like gold in return. The theory states that government should play a vital role in regulating the economy to encourage exports and discourage imports by using subsidies and taxes. According to this theory, the government should accumulate as much gold as possible, which can only be done through exports.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Explain the difference between Domestic and International Trade.

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2) Write down a short note on Mercantilism.

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1.3 ABSOLUTE ADVANTAGE THEORY

Several theories have been developed in light of the activities discussed above. These theories have evolved for more than 200 years now. Adam Smith propounded the basic theory of international trade in 1776. Adam Smith in his seminal work 'An inquiry into the Nature and Causes of the Wealth of the Nation' for the first time, provided a theoretical explanation of why trade should take place between two nations.

Adam Smith, in his book propounded the theory of absolute advantage, which states that a country should specialize in those products which it can produce efficiently, where efficiency is measured in terms of absolute labour costs. This theory assumes that there is only one factor of production: labour.

Adam Smith wrote in 'The Wealth of Nations', "If a foreign country can supply us with a commodity cheaper than we can make it, better buy it of them with some part of the produce of our industry, employed in a way in which we have some advantage".

According to Adam Smith, trade between two nations is based on absolute cost advantage. When one is more efficient than (or has an absolute advantage over) another in the production of one commodity but is less efficient than (or has an absolute disadvantage with respect to) the other nation in producing a second commodity, then both the nations can gain by each specializing in the production of the commodity of its absolute advantage and exchanging part of its output with the other nation for the commodity of its absolute disadvantage. In this process, the resources are utilized most efficiently and the output of both commodities will rise. This increase in output of both the commodities measures the Gains from specialization in production available to be divided between the two nations through trade.

In very simple words, he stated that trade would benefit both countries if country A exported the goods, which it can produce at a lower cost than country B and imported the goods, which country B can produce at a lower cost than it.

The above description of the theory can be understood better with the following example. Suppose there are two countries A and B, which produce wheat and rice with an equal amount of resources i.e. 200 labourers. Country A uses 10 labourers to produce 1 ton of wheat and 20 labourers to produce 1 ton of rice. Country B uses 25 units of labourers to produce 1 ton of wheat and 5 units of labourers to produce 1 ton of rice.

Table 1.1 Number of labourers used by Country A and Country B for producing 1 ton of Wheat and Rice

	Country A	Country B
Wheat (1 ton)	10 labourers	25 labourers
Rice (1 ton)	20 labourers	5 labourers

This is evident from Table 1.1 that country A has an absolute advantage in producing wheat as it can produce 1 ton of wheat by using fewer labourers as compared to country B. On the other hand, country B has an absolute advantage in producing rice as it can produce 1 ton of rice by employing fewer labourers in comparison to country A. Now, if there is no trade between these countries and resources (in this case there are a total of 200 labourers) are being used equally to produce wheat and rice, country A would produce 10 tons of wheat ($100/10$), and 5 tons of rice ($100/20$) and country B would produce 4 tons of wheat ($100/25$) and 20 tons of rice ($100/5$). Thus, total production without trade (i.e., autarky) is 39 tons (14 tons of wheat and 25 tons of rice).

Table 1.2 Production volume of wheat and rice without trade

	Country A (in Tons)	Country B (in Tons)
Wheat	10	4
Rice	5	20

Now, if both countries open up to trade with each other and specialize in goods in which they have an absolute advantage, the total production would be higher. If a trade takes place, Country A would produce 20 tons of wheat with 200 labourers; whereas, country B would produce 40 tons of rice with 200 labourers. Thus, total production would be 60 units (20 tons of wheat and 40 tons of rice).

Table 1.3 Production volume of wheat and rice with trade

	Country A (in Tons)	Country B (in Tons)
Wheat	20	0
Rice	0	40

It is clear from the above descriptive example that without specialization, the total production of countries was 39 tons which increased to 60 tons after specialization from trade opportunities. This is the welfare gains from trade. Therefore, the theory of absolute advantages shows that pattern of trade based on absolute cost differences would benefit both the countries.

1.4 COMPARATIVE ADVANTAGE THEORY

Elaborating on the theory of absolute advantage, David Ricardo presented his work in 1817 in his book 'Principles of Political Economy and Taxation'. In this book, he presented the law of comparative advantage. This is one of the most important and still unchallenged laws of economics with many practical applications. The Ricardian theory states that trade can be beneficial for two countries even if one country has an absolute advantage in all the products and the other country has no absolute advantage in any of the products. Ricardo argued that “...a nation, like a person, gains from the trade by exporting the goods or services in which it has its greatest comparative advantage in productivity and importing those in which it has the least comparative advantage”. A nation behaves no differently from an individual who does not attempt to produce all the commodities he needs. Rather, he produces only that commodity that he can produce most efficiently and then exchanges part of his output for the other commodities he needs or wants. The first nation would be better off provided they specialize in the production and export of those commodities in which its absolute disadvantage is smaller (this is the commodity of its comparative advantage) and import the commodity in which its absolute disadvantage is greater (this is the commodity of its comparative disadvantage).

This theory assumes that labour is the only factor of production in two countries, with zero transport cost, and no trade barriers within the countries. This theory can be understood better from the example cited in Table 1.4.

Suppose there are two countries, A and B, producing two commodities, wheat and rice, with labour as the only factor of production. Now assume that both the countries have 200 labourers and use 100 labourers to produce wheat and 100 labourers to produce rice.

Table 1.4 Production of wheat and rice by Country A and B before the trade

	Country A (in tons)	Country B (in tons)
Wheat	20	15
Rice	40	10

Table 1.4 shows that country A can produce 20 units; whereas, country B can produce 15 units of wheat by employing the same number (100) of labourers. In addition, country A can produce 40 units; whereas country B can produce 10 units of rice by employing 100 labourers.

Thus, country A has an absolute advantage in producing both items. Country A employs the same number of labourers (100 labourers in the production of each item) in producing rice and wheat; however, rice production is higher than wheat production. This reveals that country A has a comparative advantage in producing rice. Similarly, country B also employs the same number of labourers (100 labourers in producing each good) in manufacturing wheat and rice; but, its wheat production is more than rice. It indicates that country B has a comparative advantage in producing wheat.

For example, country A has decided to produce 60 units of rice by employing 150 labourers. It uses 50 labourers to produce 10 units of wheat. On the other hand, country B has decided to use all the 200 labourers to produce 30 units of wheat and stop the production of rice. In this situation, country A exchanges 14 units of rice with 14 units of wheat produced by country B. (see Table 1.5).

Table 1.5 Production after specialization

	Country A	Country B
Wheat	10	30
Rice	60	0

Table 1.6 Situation after the trade takes place

	Country A	Country B
Wheat	24	16
Rice	46	14

It can be understood from Table 1.6 that both countries have benefitted or gained from trade. Before the trade, country A had only 20 units of wheat and 40 units of rice; after the trade, country A had 24 units of wheat and 46 units of rice.

At the same time, country B has 15 units of wheat and 10 units of rice before the trade, while it has 16 units of wheat and 14 units of rice after the trade. Therefore, comparative advantage theory clarifies that trade can create benefits for both participating countries even if one country has an absolute advantage in the production of both commodities.

Thus, it is clear from the above theories that the mercantilists believed that one nation could gain only at the expense of another nation and advocated strict government control and regulations of all economic activities. Adam Smith believed that all nations would gain from free trade and strongly advocated a policy of laissez-faire (i.e., as little government interference with the economic system as possible). Free trade would cause world resources to be utilized most efficiently and would maximize world welfare. There were to be only a few exceptions to this policy of laissez-faire and free trade. One of these was the protection of industries important for national defence.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Explain the central proposition of the absolute advantage theory and comparative advantage theory.

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- 2) Differentiate between Adam Smith and Ricardo's theory of international trade.

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- 3) With reference to the Ricardian theory of international trade, explain how nations gain from trade.

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1.5 HECKSCHER -OHLIN THEORY

In the preceding section, we have seen that the difference in relative commodity prices between two nations is the reason for their comparative advantage, which forms the basis for mutually beneficial trade. Then we explained the reason, or cause, for the difference in relative commodity prices and comparative advantage between the two nations. The second way we extend our trade model is to analyze the effect of international trade on the earnings of factors of production in the two trading nations. That is, we want to examine the effect of international trade on the earnings of labour as well as on international differences in earnings. These two important questions were left largely unanswered by Smith, Ricardo, and Mill. According to classical economists, the comparative advantage was based on the difference in the productivity of labour among nations, but they did not explain such a difference in productivity, except for possible differences in climate. The Heckscher–Ohlin theory goes much beyond that by extending the trade model of the previous sections to examine the basis for comparative advantage and the effect that trade has on factor earnings in the two nations. (Salvatore 2013)

Bertil Ohlin has advocated the Modern Theory of international trade. Ohlin has drawn his ideas from Eli Heckscher's General Equilibrium Analysis. Hence it is also known as Heckscher Ohlin (H-O) Theory.

Heckscher-Ohlin theory begins where the Ricardian theory of international trade ends. The Ricardian theory states that the basis of international trade is the comparative or relative cost differences. But he did not explain how this comparative cost difference emerges. Ohlin's theory explains the real cause of this difference.

Ohlin states that trade takes place due to the different relative prices of different goods in different countries. The difference in the relative price of a commodity is due to differences in the relative costs and factor prices in different countries.

According to the H-O theory, the differences in factor prices are due to differences in factor endowments in different countries. The trade pattern is explained in terms of differences in relative factor endowments i.e., the relative resource supply in an economy. Ohlin's theory is also described as the factor endowment theory. Ohlin's theory is usually expounded in terms of a two-factor model with labour and capital as the two factors of endowments. Trade occurs due to the differences in factor endowments. Some countries have plenty of capital, while others have an abundance of labour. The Heckscher-Ohlin theorem postulates that countries rich in labour will export labour-intensive goods and countries with plenty of capital will export capital-intensive products.

Heckscher-Ohlin's theory explains the modern approach to international trade based on the following assumptions: -

- 1) There are two countries involved, countries A and B
- 2) There are two factors of production, labour and capital.
- 3) There are two goods, X and Y, of which X is labour-intensive, and Y is capital-intensive.
- 4) Country A is a labour-abundant country B is capital-rich.
- 5) There is perfect competition in both the commodity and factor markets.
- 6) All production functions are homogeneous of the first degree. Hence there are constant returns to the scale.
- 7) There are no transport costs or other impediments to trade.
- 8) Demand conditions are identical in both countries.

These assumptions have been made to explain the meaning of comparative price advantage or relative price difference and to deduce the major propositions of the factor endowment theory. Given these assumptions, Ohlin's thesis contends that, country exports goods which use relatively a greater proportion of its relatively abundant and thus cheap factors. It is implied that trade occurs because differences in relative commodity prices are caused by differences in relative factor prices (thus a comparative advantage) because of differences in the factor endowments among the countries.